Automating firewall rules deployment in a highly regulated environment

NETWORK AUTOMATION FORUM

Ørsted

AUTOCON 5

THE NETWORK AUTOMATION CONFERENCE

About me

- 32 years old
- Portuguese, moved to Sweden, working in Denmark. Met wife while studying Japanese
- Speaks none of those languages. Barely even Portuguese and English
- Studied software engineering. Didn’t feel it. Moved to network and security. Missed programming. Here we are

Now a bit more professionally

- Largest offshore wind developer in the world
- Biggest energy company in Denmark
- Over 30 offshore windfarms in operation
- Currently developing 8+ with a combined 8 GW

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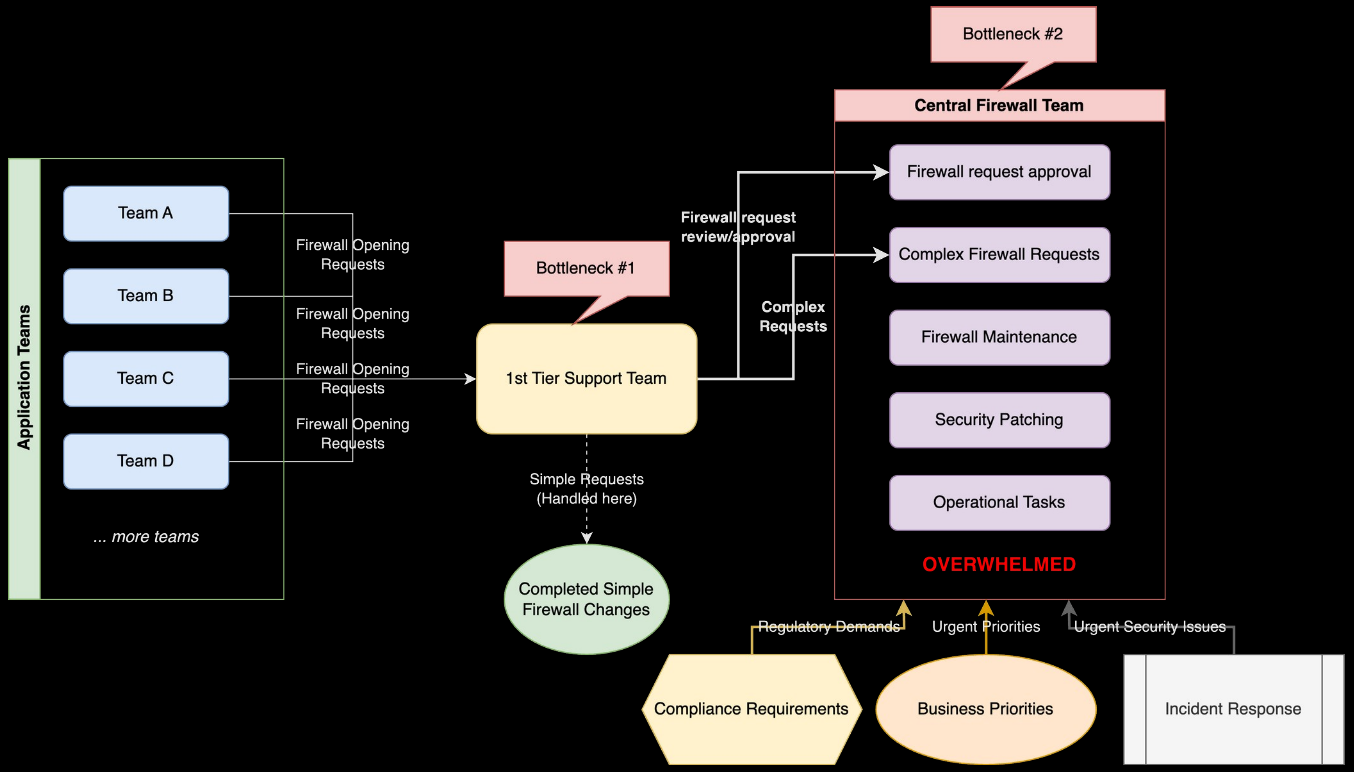
A tale as old as time

The beauty of firewall requests



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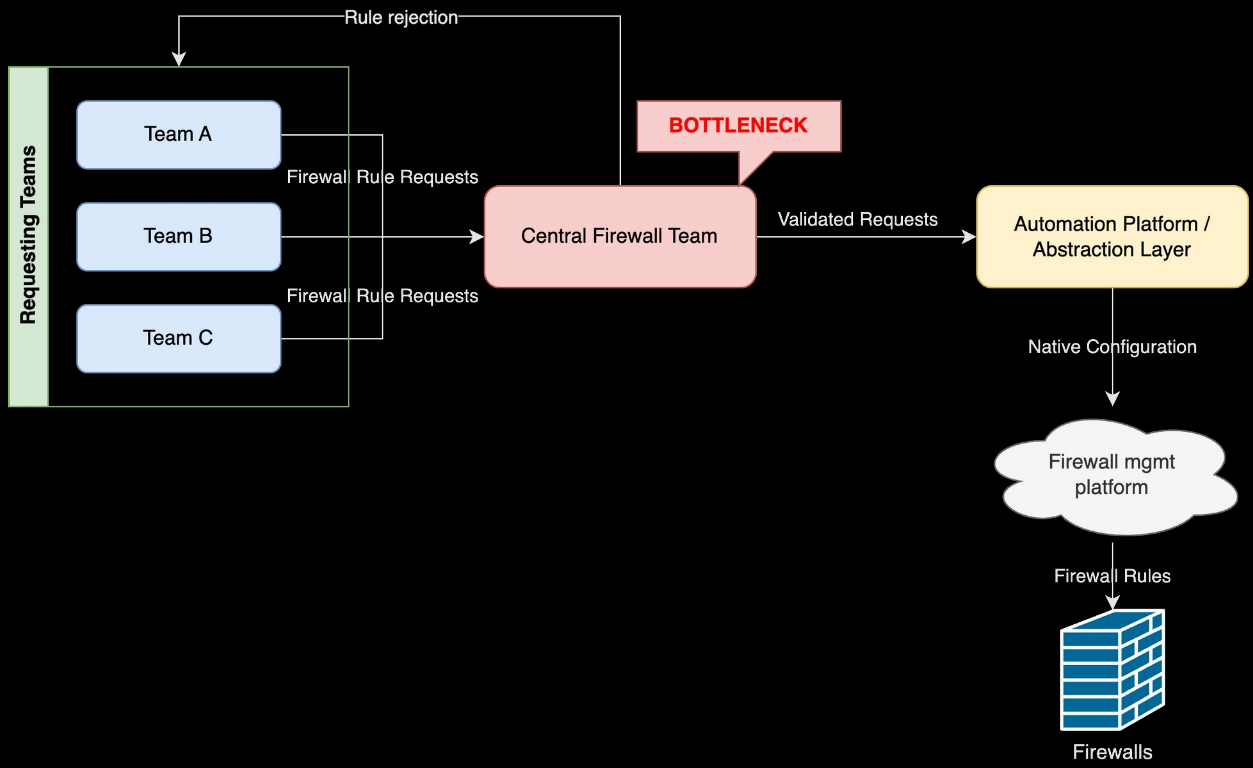
Automation?!
What options do we have?

Option 1 - Automate the deployment?



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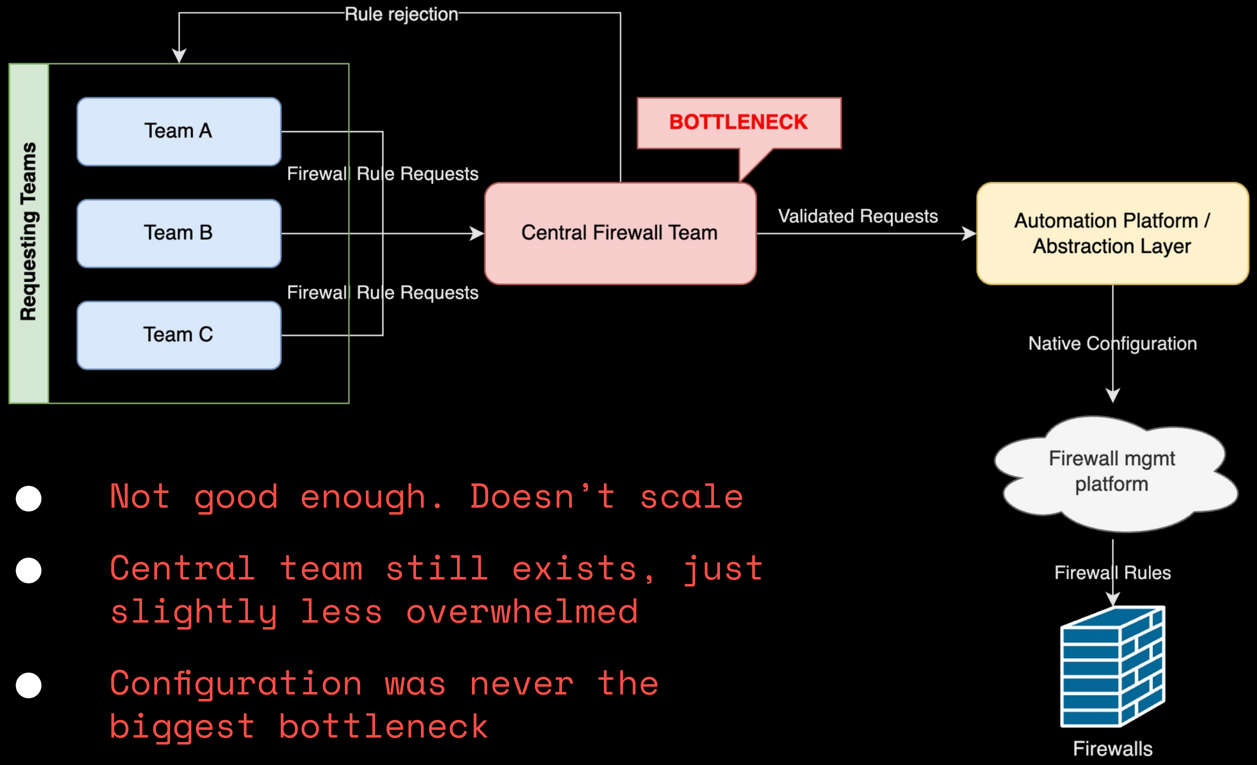
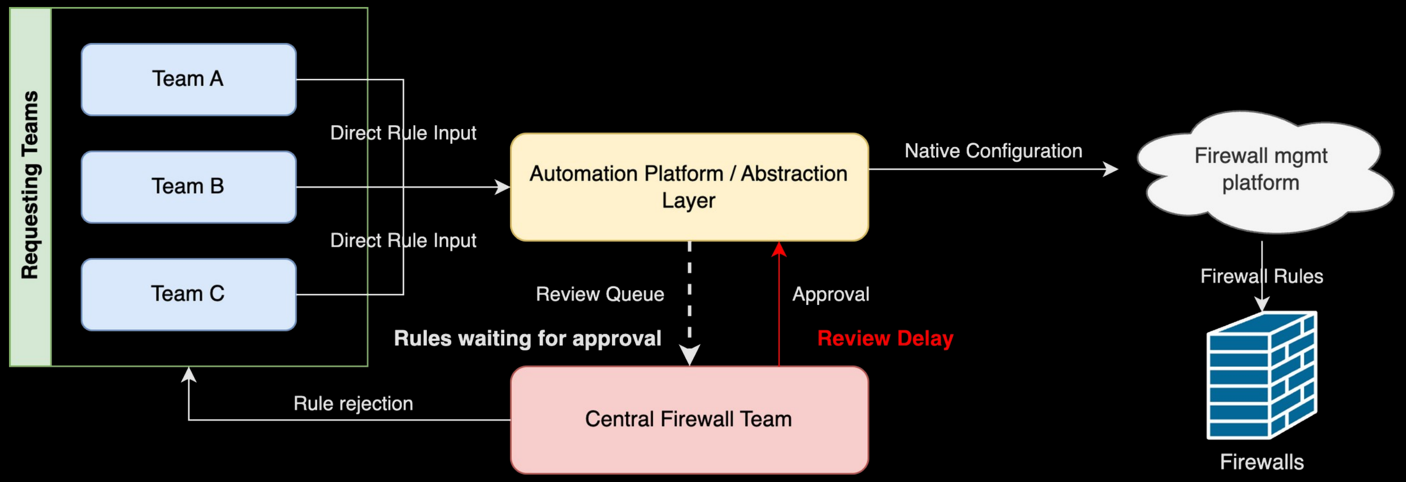
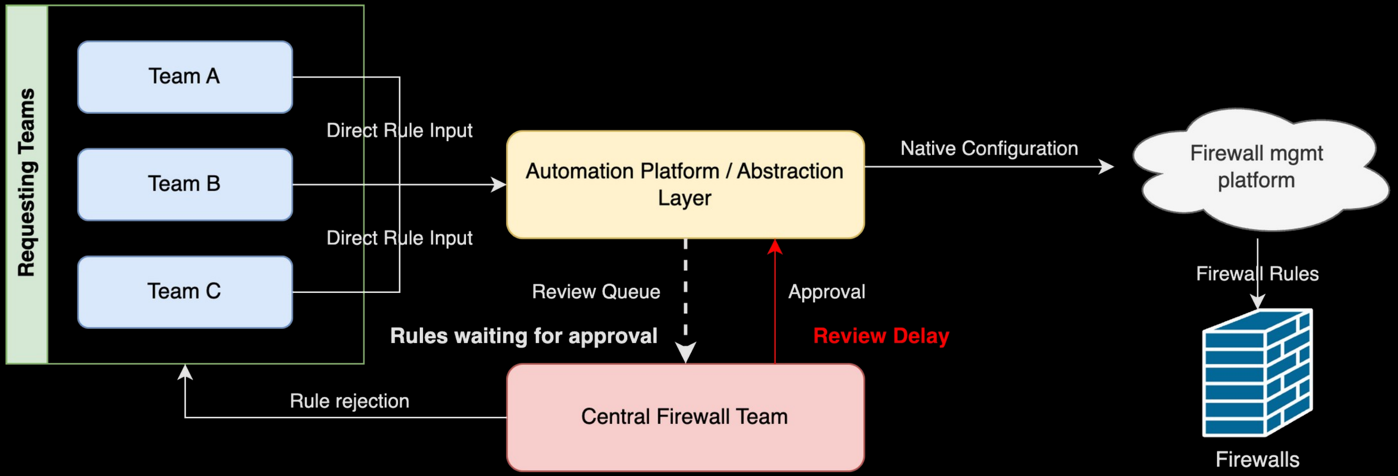


- Not good enough. Doesn't scale
- Central team still exists, just slightly less overwhelmed
- Configuration was never the biggest bottleneck

Option 2 - Automate requests and deployment?

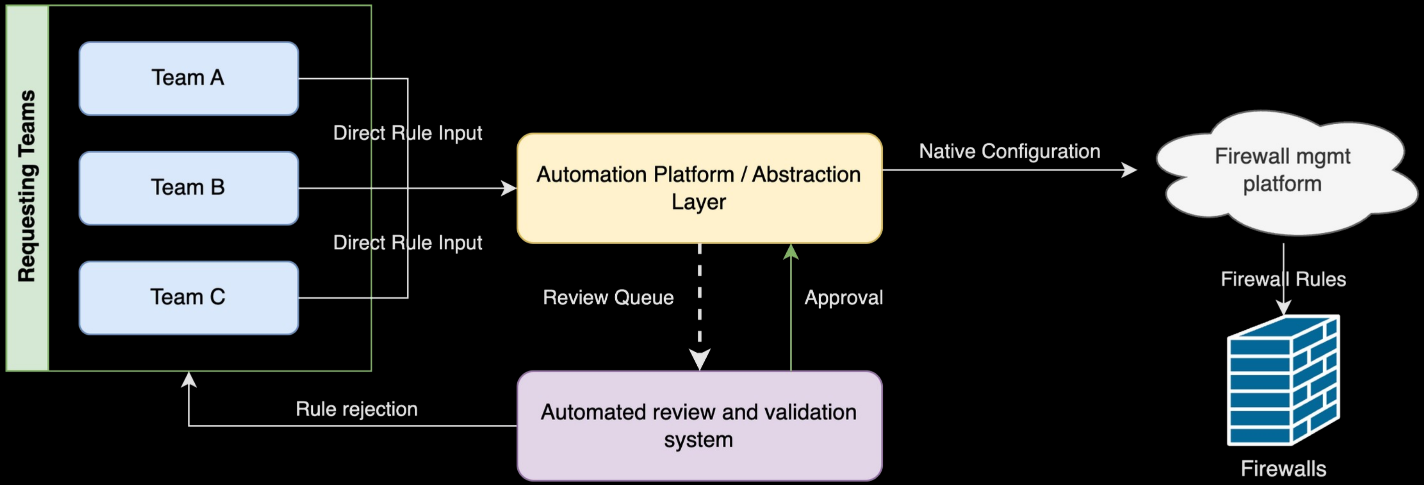


Option 2 - Automate requests and deployment?



- Almost there, but still not scalable
- Central team still exists, spends several hours reviewing requests
- Requests go back and forth until approved. Customers still angry

Option 3 - Automate requests, validation and deployment!



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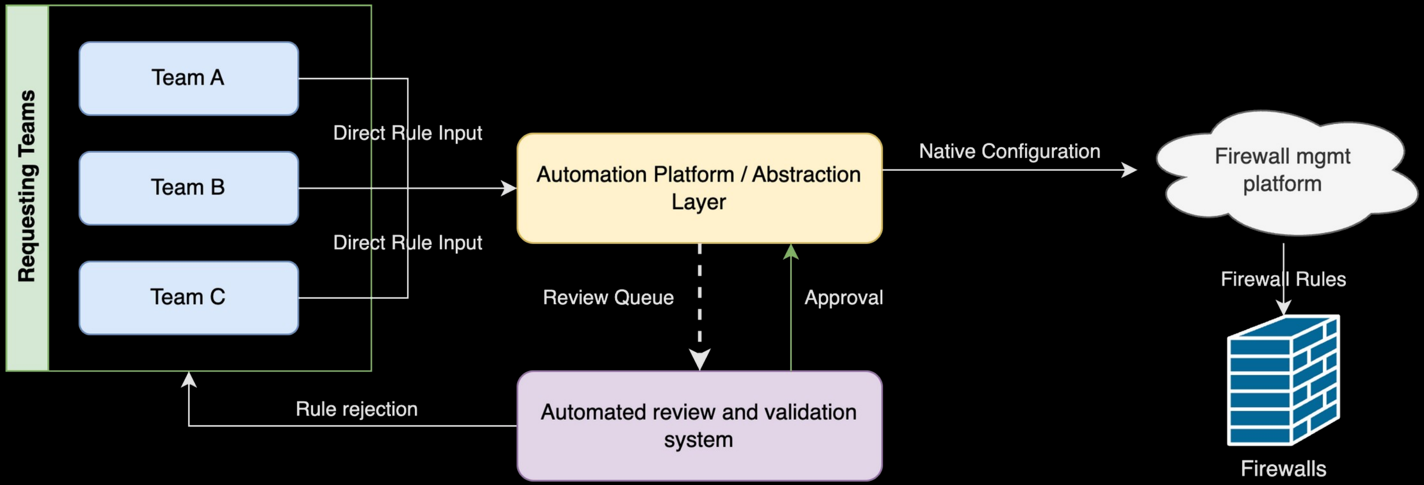

We have a dream!

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Our journey

The requirements. Non-negotiables

- Version control and ability to audit every single firewall rule and revert if necessary
- An automated rule validation system
- A platform to host and run the product. Ability to modify, extend, customize our solution
- Reduce dependencies on 3rd party vendors and closed source products
- Strong community behind whatever platforms we choose
- Abstract complexity from our customers. As user-friendly as possible

The requirements → Tools

- Version control and ability to audit every single firewall rule and revert if necessary → Git
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The requirements → Tools

- Version control and ability to audit every single firewall rule and revert if necessary → Git
- An automated rule validation system → N/A – Developed from scratch. Python-based
- A platform to host and run the product. Ability to modify, extend, customize our solution
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Git

N/A - Developed from scratch. Python-based

Nautobot
Terraform
Ansible

The requirements → Tools

- Version control and ability to audit every single firewall rule and revert if necessary → Git
- An automated rule validation system → N/A - Developed from scratch. Python-based
- A platform to host and run the product. Ability to modify, extend, customize our solution → Nautobot
- Reduce dependencies on 3rd party vendors and closed source products → Terraform
- Strong community behind whatever platforms we choose → Ansible
- Abstract complexity from our customers. As user-friendly as possible → YAML-based structure

Validation and compliance!

Cool, what guidelines do we have to follow?

Guidelines?
Do you mean SharePoint?

- Security principles provided in SharePoint documents
- Regulations across 100+ pages of Word documents
- Gut-feelings via Teams chats and meetings from IT security teams on what rules can be allowed

The power of a source of truth

- A source of truth which can you interact with both manually and programmatically is a <u>requirement</u> for an effective validation strategy

- Translate as much data as possible into code and custom models. No more docs and sheets. Only policy-as-code.

- <u>**Your validation will only ever be as good as the quality of your data**</u>. Enforce data constraints and sanitization!

Example security principle: Before

IT Security:

“We approve firewall rules between ***Onprem IT*** *and* ***Azure*** *using* ***well-known and encrypted protocols****”*

Example security principle: After

>>> Onprem IT

May 15, 2026, 12:50 p.m. | 7 minutes ago

Security Zone | Advanced | Contacts | Change Log

Security Zone	
Name	Onprem IT
Description	
Trust Level	TL2

Relationships	
EPGs	—
Prefixes	—
VRFs	2 VRFs

1. Build custom models for business logic and abstract security concepts

2. Relate these models to real network constructs

>>> Azure

May 15, 2026, 12:50 p.m. | 6 minutes ago

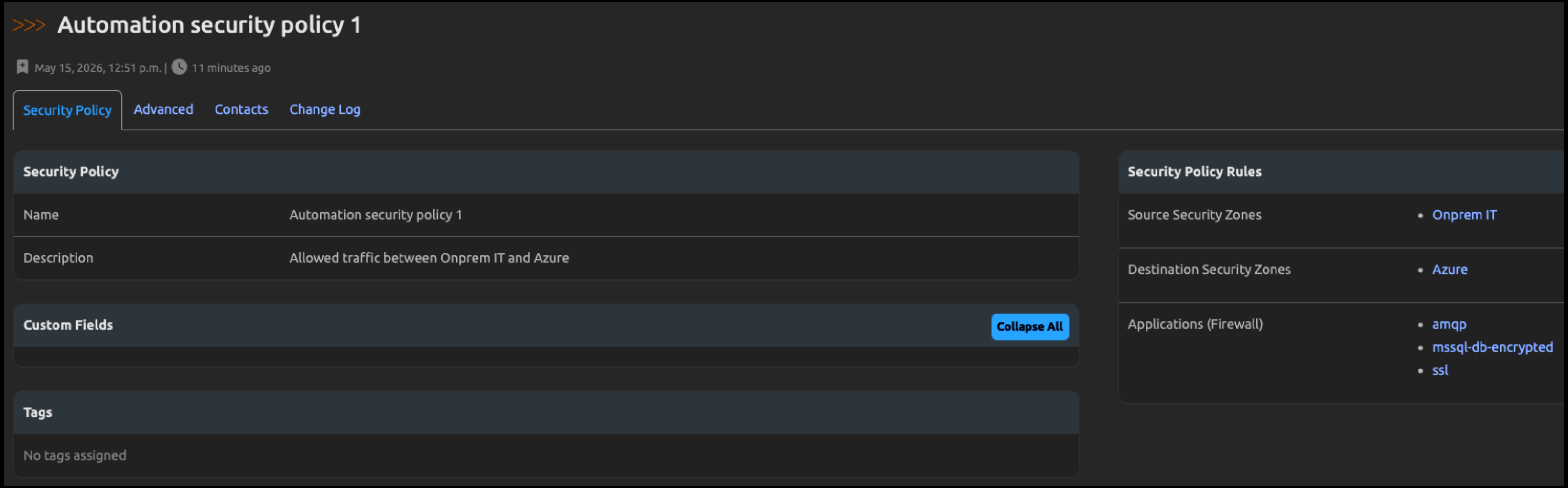
Security Zone | Advanced | Contacts | Change Log

Security Zone	
Name	Azure
Description	
Trust Level	TL2

Relationships	
EPGs	—
Prefixes	2 prefixes
VRFs	—

Relationship	Source	Destination
Security Zone to Prefix Relationship	Azure	192.168.150.0/27
Security Zone to VRF Relationship	Onprem IT	Global: (some-onprem-VRF)
Security Zone to VRF Relationship	Onprem IT	Global: (some-other-onprem-vrf)
Security Zone to Prefix Relationship	Azure	192.168.44.0/22

Example security principle: After



3. Build the policy connecting all the models!

Example security principle: After


```
HTTP 200 OK
API-Version: 2.4
Allow: GET, PUT, PATCH, DELETE, HEAD, OPTIONS
Content-Type: application/json
Vary: Accept

{
    "id": "004228f5-91dc-45ff-8f25-48df0bf6331a",
    "object_type": "nautobot_orsted.securitypolicy",
    "display": "Automation security policy 1",
    "url": "http://localhost:8080/api/plugins/orsted/securitypolicies/004228f5-91dc-45ff-8f25-48df0bf6331a/",
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    "name": "Automation security policy 1",
    "description": "Allowed traffic between Onprem IT and Azure",
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            "object_type": "nautobot_orsted.securityzone",
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            "name": "Onprem IT",
            "description": "",
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                "url": "http://localhost:8080/api/plugins/orsted/tls/422fffa1-a2e9-4cdf-881c-de3c9d9c3d28/"
            },
            "created": "2026-05-15T12:50:04.161395Z",
            "last_updated": "2026-05-15T12:50:04.161404Z",
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            "custom_fields": {}
        }
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            "url": "http://localhost:8080/api/plugins/orsted/securityzones/3ef199e9-3b89-418b-bd82-32de1d9213fe/",
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                "url": "http://localhost:8080/api/plugins/orsted/tls/422fffa1-a2e9-4cdf-881c-de3c9d9c3d28/"
            },
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            "last_updated": "2026-05-15T12:50:39.499916Z",
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            "custom_fields": {}
        }
    ],
    "applications": [
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            "object_type": "nautobot_firewall_models.applicationobject",
            "display": "amqp",
            "url": "http://localhost:8080/api/plugins/firewall/application-object/5b5d91da-6d5c-44c8-a218-a6c28d15b800/",
            "natural_slug": "amqp_5b5d",
            "description": "",
            "category": "",
            "subcategory": "",
            "technology": "",
            "risk": null,
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            "name": "amqp",
            "default_ip_protocol": "",
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                "object_type": "extras.status",
                "url": "http://localhost:8080/api/extras/statuses/df718d8c-85cf-49f0-91b9-73dbc585d2fa/"
```
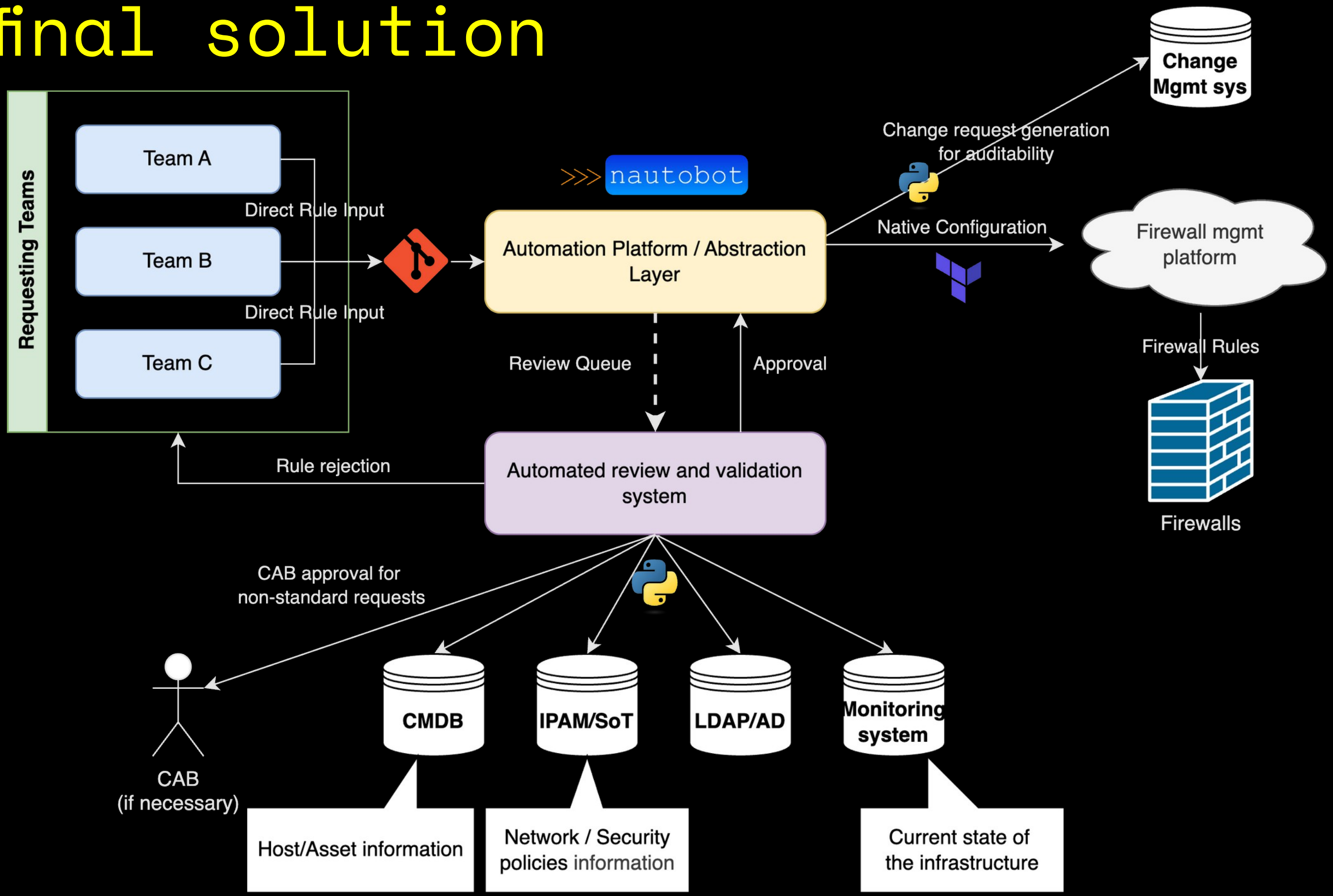


4. Expose it to humans via a GUI and machines via an API!

Now glue everything together

The final solution

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Takeaways and statistics

Some numbers as of 2026

- 40+ teams onboarded
- 150+ active users
- 3000+ failed validation attempts
- 3500+ successful deployments (rule additions, deletions, changes)

- Number of **yearly firewall requests** **<u>increased by ~250%</u>**
- **Automated requests** up from 0% to **<u>80%</u>** (to be 100% soon)
- **Lead time** for firewall requests handled by humans **<u>down by 88%</u>**

Key takeaways and advice

1. **Data is everything**. A solid data foundation goes a long way
2. **Involve your customers and security** from the start
3. **Treat your automation as a real product**. Proper name, versioning, releases, changelogs and support channels
4. **Customers start caring** about firewall rules if they own them. They even start to optimize and clean-up after themselves!
5. **Number of requests increases dramatically**. The hidden cost of an inefficient process becomes obvious after automating it


Questions?
Catch me out there!